Test plan: SkyPermit API

Version: 3 · **Date:** 2026-10-18 · **Based on:** SkyPermit API contract; open questions list

Objective and scope

Confirm that the SkyPermit API works as its contract defines: who can do what, how flight requests move from draft to a reviewer's decision, and how refusals are explained. Success means users can rely on the API to protect each operator's data and handle every flight request correctly.

Test schedule

Phase	Dates	Milestone
Test start	TBD	Entry criteria met
Test execution	TBD	All planned test cases run
Test end / release	TBD	Exit criteria met

Entry criteria

- The test environment with the API is available.
- Test data, including the zone catalogue, is ready.

Functionality to be tested

- **Accounts:** register, log in, log out. New operators always start at the lowest authorisation level.
- **Data privacy:** operators see and change only their own data; visitors are refused; reviewers can view all data.
- **Registry:** pilots and aircraft (with calculated pilot currency and weight class), and the zone catalogue.
- **Flight requests:** create, submit, cancel within the deadline, and search by status, zone and dates.
- **Submission checks:** pilot, aircraft, zone and time-window checks run in a fixed order and stop at the first failure. Each refusal names the failed check, its rule and a clear message.
- **"Check before submitting":** gives the same result as a real submission and never changes the request.
- **Reviewer decisions:** approve, or reject with a required note. Every decision is recorded.
- **Suspended operators:** blocked from making changes, tested against the provisional response.

Functionality not to be tested

- **User interface:** not part of API testing.
- **Individual check rules:** pilot ratings, weight, altitude and curfew rules are out of scope; only how the checks behave is tested.
- **Creating or editing zones:** the API offers no way to do it.
- **Raising an operator's authorisation level:** an administrator does it outside the API.
- **Reviewer login:** doesn't exist by design; reviewers identify themselves with a label only.

Test approach

- **Method:** manual testing of the API against the contract, using test cases derived from it.
- **Access:** each action tried as a visitor, the owning operator, another operator and a reviewer.
- **Statuses:** each flight request action tried from allowed and not-allowed statuses (state transition testing).
- **Inputs:** valid, missing and malformed data (equivalence partitioning).
- **Consistency:** "check before submitting" compared with a real submission on the same request.

Risks covered

Risk	Impact if not caught	Coverage
Another operator's data exposed	Data breach, loss of trust	Access tests for every role
Request approved or moved without review	Flights authorised outside the process	Status tests for every action
Pre-submission check alters the request	Operators misled	Side-by-side comparison
Refusals don't explain what failed	Requests can't be fixed or traced	Every refusal checked
Suspended-operator response undecided	Tests may need redoing	Provisional tested; owner decides

Exit criteria

- No open blocker or critical defects.
- All positive test cases have passed.